



Chapter Notes: Nature of Matter: Elements, Compounds, and Mixtures

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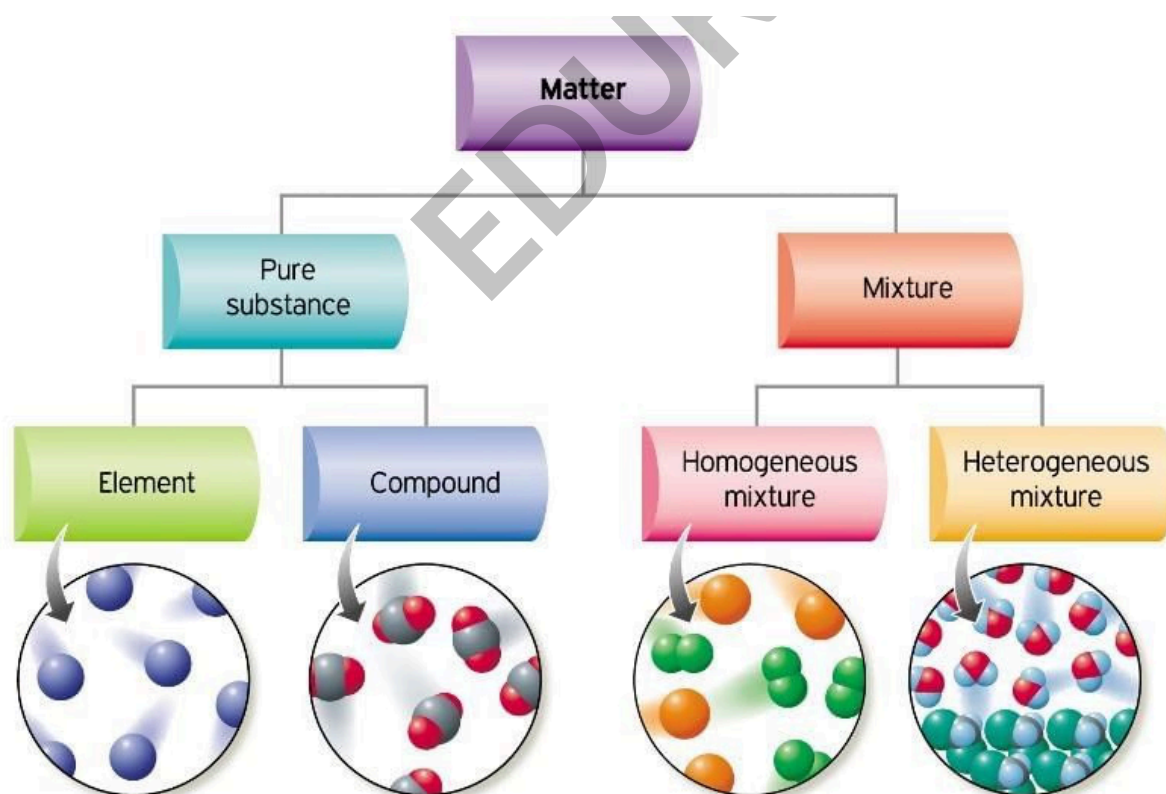
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Introduction

What is everything around you really made of?

From the air you breathe to your food, your clothes, and even your school books-**almost everything is matter made of tiny particles.**

Most everyday materials are not single pure substances; they are formed by combining different substances in various ways so that each retains or changes its properties depending on how they are combined.

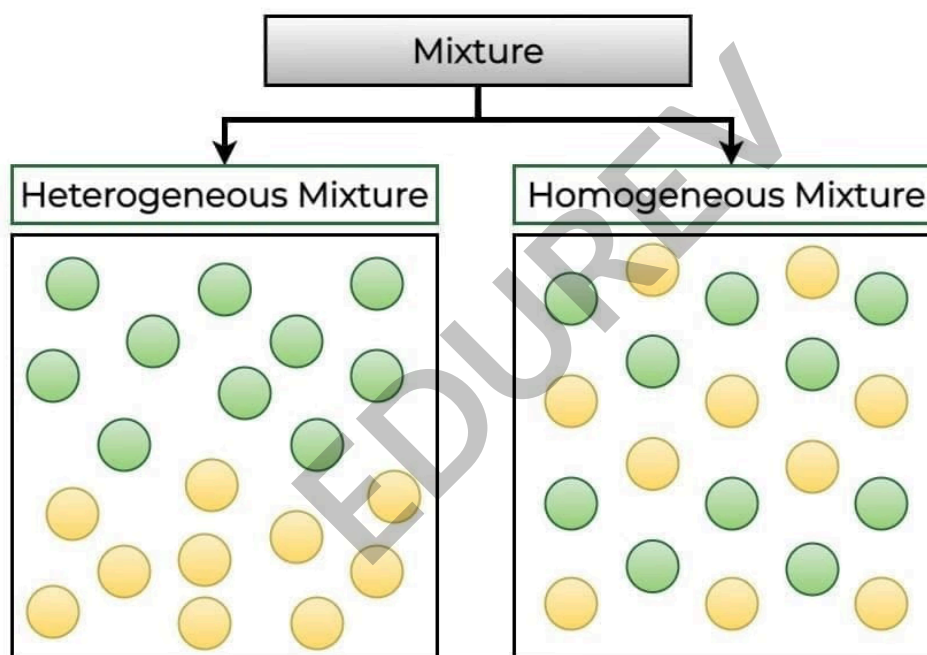


In this chapter you will study **elements, compounds, and mixtures**. You will learn how they differ, how mixtures can be separated, how compounds are formed and broken, and how this knowledge is applied in daily life and industry.

What Are Mixtures?

A **mixture** is formed when two or more substances are combined such that **each substance keeps its own properties** and **they do not react chemically** with each other. The individual substances in a mixture are called its **components**.

Types of Mixtures



Mixtures are classified as:

Heterogeneous (non-uniform) mixtures: The different components can be seen and often separated by simple physical methods. Examples: poha, sprout salad, sand and salt, trail mix, vegetable soup.

Homogeneous (uniform) mixtures: The components are mixed so well that they appear as a single substance and cannot be seen separately. Examples: sugar dissolved in water, saltwater, air, and many alloys like stainless steel.

Mixtures are common: Most foods, beverages and many materials we use are mixtures.

Properties: The properties of a mixture depend on the proportion of its components and can vary from sample to sample.

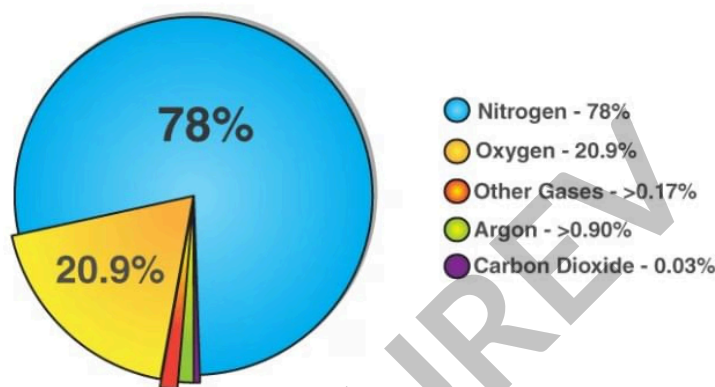
Scientific Heritage: Alloys

Alloy (Mishraloha): A mixture of two or more metals (or a metal and another element) made to get improved properties. Examples: **bronze** (copper + tin), **brass** (copper + zinc), **stainless steel** (iron + nickel + chromium + carbon).

Alloys are usually homogeneous mixtures and are important in construction, tools and appliances because they combine desirable properties of their components.

Is Air a Mixture?

Air is a homogeneous mixture of gases, mainly nitrogen, oxygen, argon, carbon dioxide and water vapour.



Components of Air

Why air is a mixture:

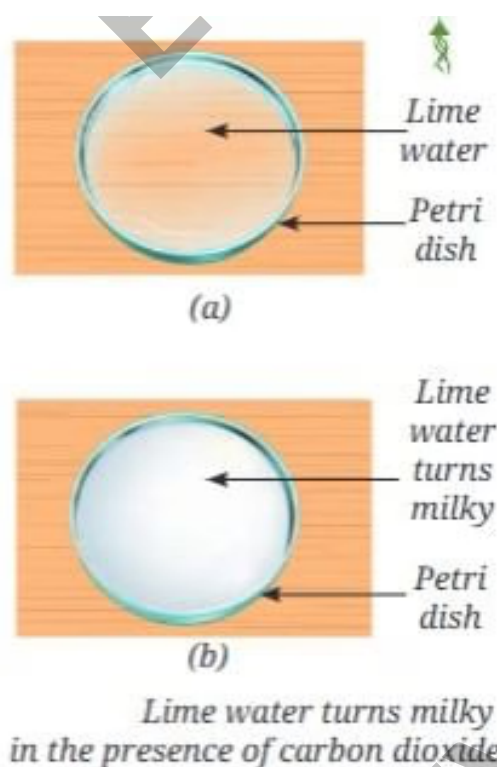
The gases in air are not chemically bonded; each gas keeps its own properties.

Air is uniform in composition at a given place and time (homogeneous mixture).

You cannot separate the gases by sight; separation needs physical methods such as fractional distillation of liquid air.

Testing for Carbon Dioxide in Air

Prepare lime water by adding calcium oxide (quicklime) to water to form a solution of calcium hydroxide. Expose lime water to air.



The clear solution turns **milky** on exposure to air.

This happens because carbon dioxide in the air reacts with calcium hydroxide to form insoluble calcium carbonate, which appears milky.

Conclusion: The air around us contains carbon dioxide.

Dust, Particulates and Air Pollution

Leave a clean black sheet of paper near an open window for a few hours and observe dust settling on it; examine with a magnifying glass if available.

Dust particles and soot are present in air and are classified as **particulate pollutants**.

Major air pollutants include particulate matter, carbon monoxide, ozone, nitrogen dioxide and sulphur dioxide. Air quality is often described using the **Air Quality Index (AQI)**.

Types of Mixtures by Physical State

Mixtures can be formed by combining substances in any states of matter: solid in solid, solid in liquid, liquid in liquid, gas in gas, gas in liquid, and so on. Examples include alloys (solid-solid), salt dissolved in water (solid-liquid) and air (gas-gas).

S.No.	Mixture-type	Examples	Uniform or non-uniform
1.	Gas and gas	Air	Uniform
2.	Gas and liquid	Aerated water (soda water) Oxygen dissolved in water	Uniform Uniform
3.	Solid and gas	Carbon particles in air	Non-uniform
4.	Liquid and liquid	Acetic acid in water (vinegar) Oil and water	Uniform Non-uniform
5.	Solid and liquid	Sand and water Seawater	Non-uniform Uniform
6.	Solid and solid	Baking powder (baking soda and tartaric acid) Alloys	Non-uniform Uniform

Why separate mixtures?

In daily life: To obtain a wanted component (e.g., remove stones from rice, separate tea leaves from tea).

In science and industry: To obtain pure substances for research, medicines, analysis or further processing.

What Are Pure Substances?

Everyday meaning:

On food packets, "pure" usually means unadulterated-without cheaper or harmful substances added.

Adulteration is the illegal mixing of inferior or harmful substances and reduces quality and safety.

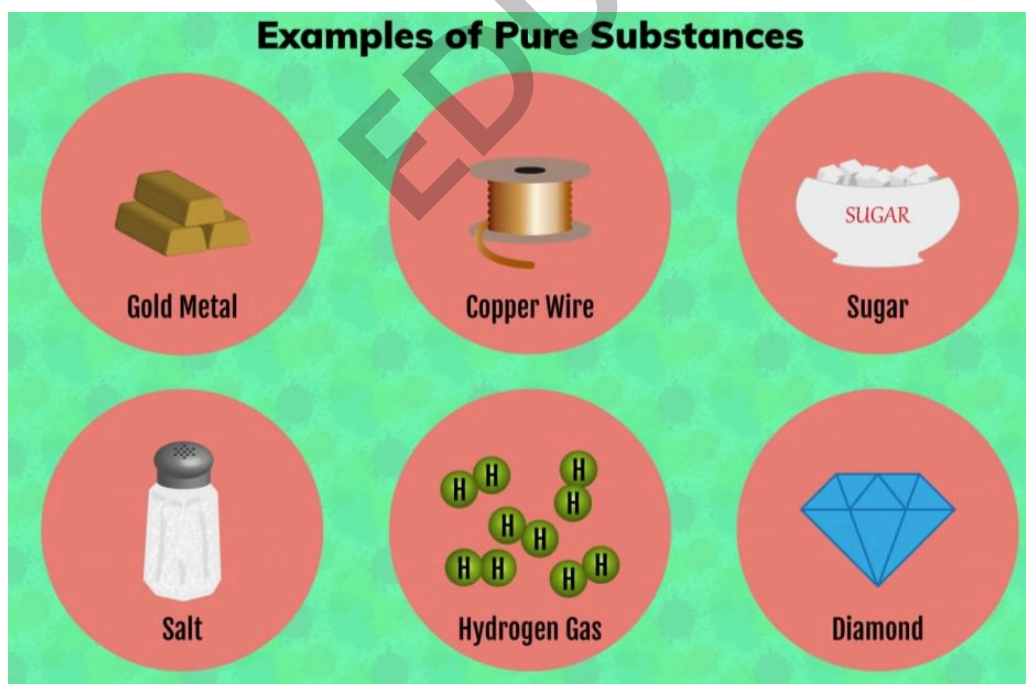
Scientific definition of a pure substance

Contains only one kind of matter (one type of particle) throughout.

Examples: Chemically pure sugar, distilled water, baking soda (if chemically pure), pure iron, pure gold (24K).

A **pure substance cannot be separated into other substances by physical processes** such as filtering, sieving or simple evaporation.

Mixtures like milk, soil and packed juice are not pure substances.



Q. Classify the following as mixture or pure substance: Milk, Packaged fruit juice, Baking soda, Sugar, Soil.

Milk: Mixture (contains water, fats, proteins, minerals).

Packaged fruit juice: Mixture (water, sugars, flavours, vitamins and often added preservatives).

Baking soda: Pure substance (if chemically pure - sodium bicarbonate only).

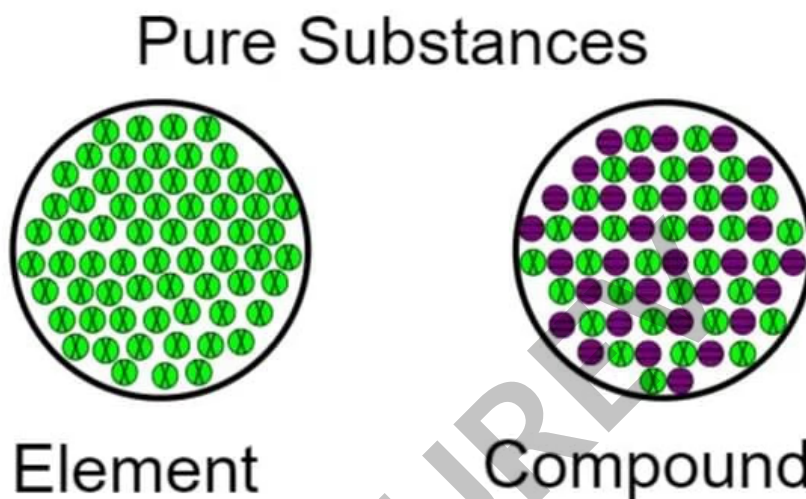
Sugar: Pure substance (if it is pure sucrose).

Soil: Mixture (sand, clay, organic matter, water and air).

In science, "pure" means only **one substance present**-the same kind of particle everywhere in the sample.

Types of Pure Substances

Pure substances are divided mainly into **elements** and **compounds**.



Physical changes

Physical changes (melting, freezing, evaporation) change the state of a pure substance but not its chemical identity. For example, ice, water and steam are all the same substance-water.

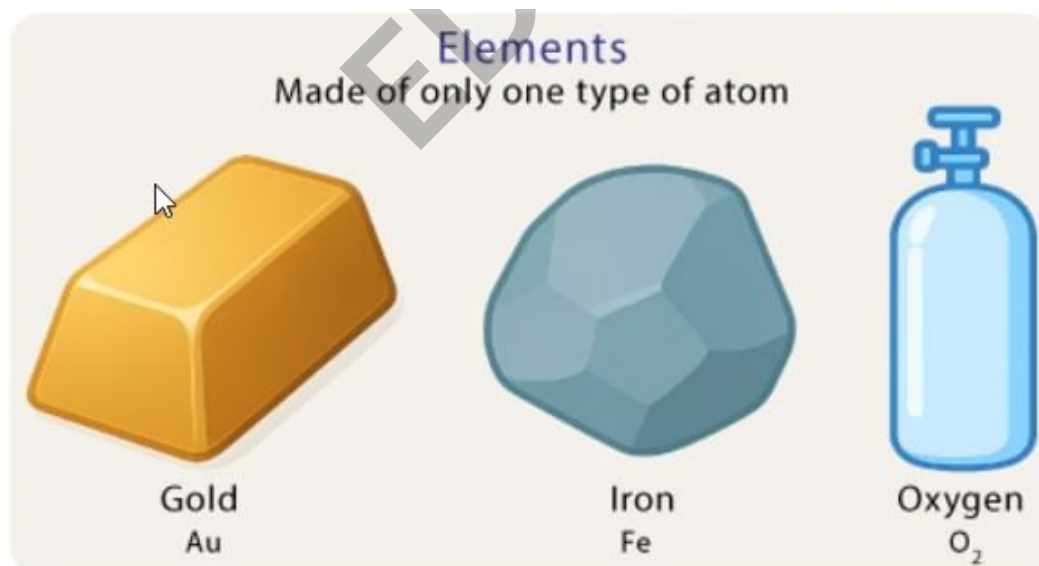
Chemical decomposition-example of water

When electricity is passed through water (electrolysis), water decomposes into two gases: hydrogen and oxygen.

This chemical change shows water is a compound made of two elements.

Elements

Elements are pure substances made of only one type of atom.



Elements cannot be broken into simpler substances by chemical means.

Each element is made of identical atoms.

Examples: Hydrogen, oxygen, gold, silver, sulfur, carbon.

Molecules and forms

Many elements exist as molecules (atoms joined together).



Metals (e.g., gold, iron) are generally shiny, malleable and good conductors of heat and electricity.

Non-metals (e.g., oxygen, sulfur) are often poor conductors and can be gases, liquids or brittle solids.

Metalloids (e.g., silicon, boron) have properties between metals and non-metals; these are studied in higher classes.

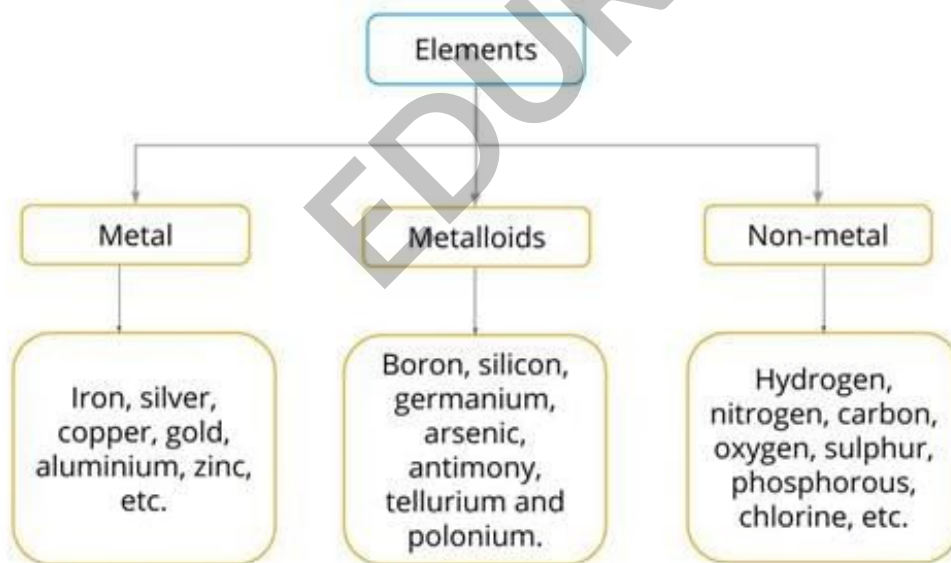
Key facts about elements

The total known elements are 118.

Most elements are solids at room temperature; eleven are gases (for example oxygen, helium, nitrogen).

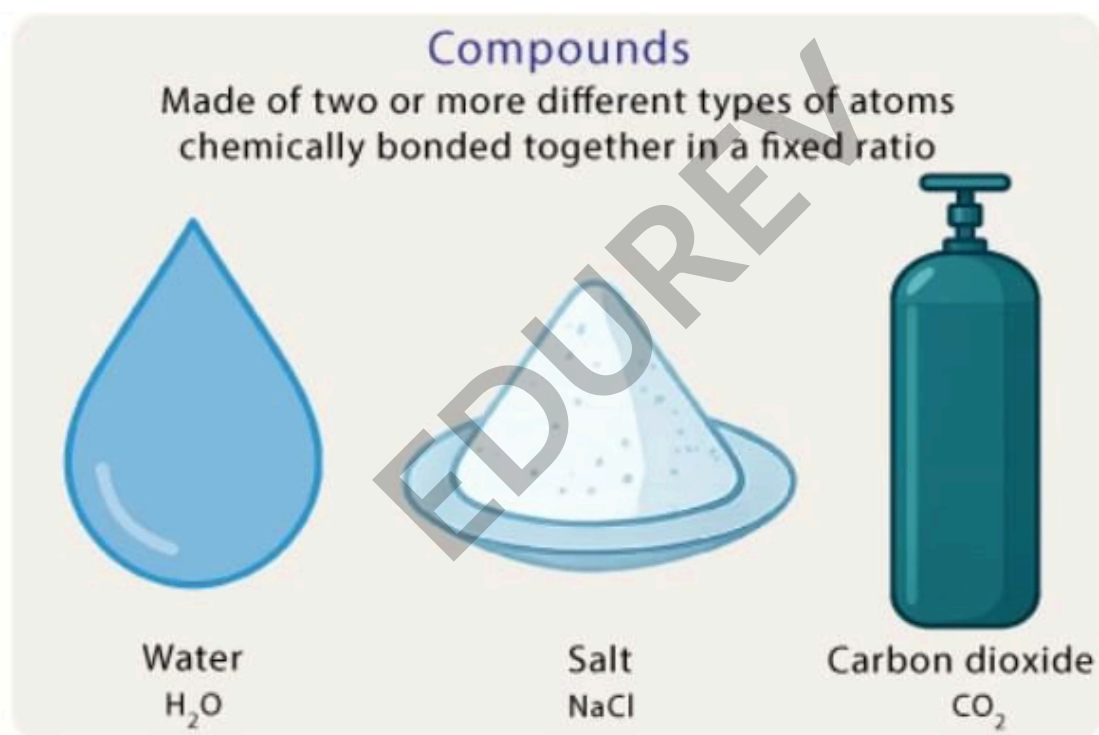
Two elements are liquids at room temperature: **mercury** (a metal) and **bromine** (a non-metal).

Some elements such as gallium and caesium are solids that melt at temperatures close to 30°C . Their melting behaviour is a special case.



Compounds

Compounds are pure substances formed by chemical combination of two or more elements in a fixed ratio.



Key properties of compounds

Compounds form in **fixed ratios** of elements. For example, water is always formed by two hydrogen atoms combined with one oxygen atom, **ratio (2:1)**

Compounds have properties different from their constituent elements. For example, hydrogen (a flammable gas) and oxygen (which supports combustion) combine to form water (which extinguishes fire).

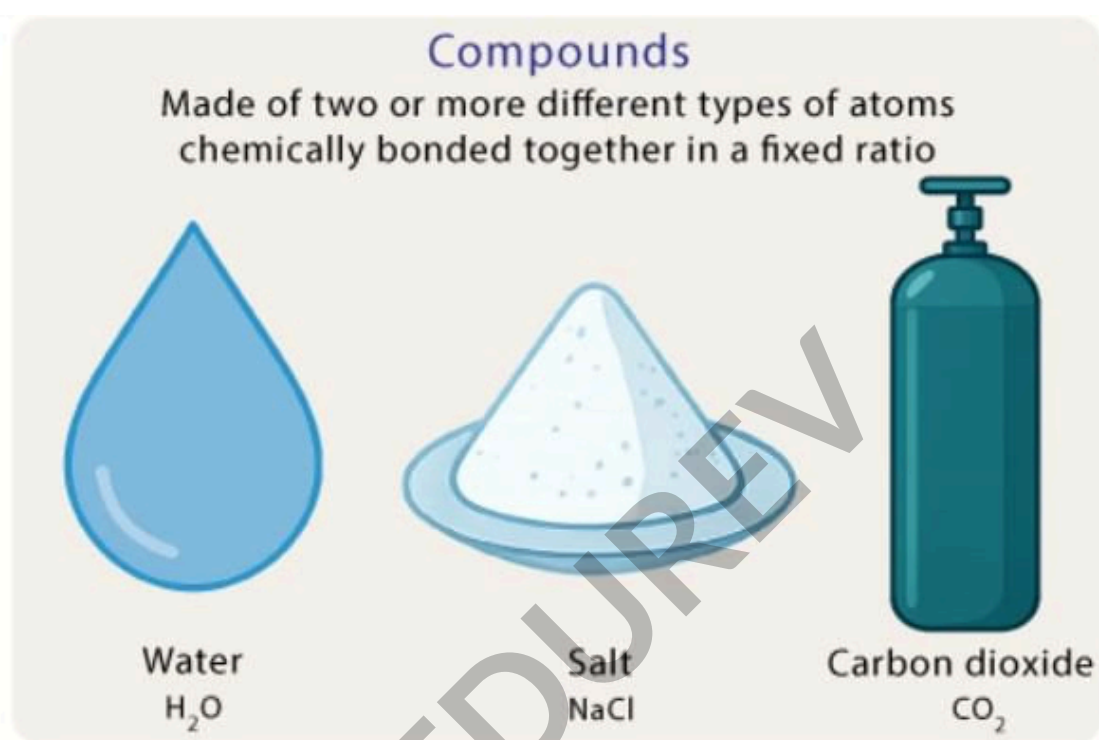
Examples of compounds

Water: a compound of hydrogen and oxygen.

Sodium chloride (table salt): a compound of sodium (a soft metal) and chlorine (a poisonous gas) in a 1:1 ratio. Salt is safe to use; separating sodium and chlorine requires chemical methods.

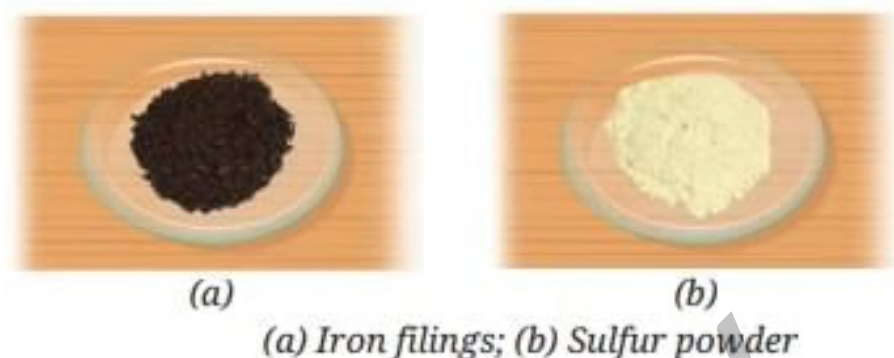
Sugar: a compound of carbon, hydrogen and oxygen. On strong heating sugar decomposes (carbonisation) rather than breaking into elemental carbon and water easily.

Interesting fact: Modern mobile phones use more than 45 elements including aluminium, copper, silicon, cobalt, lithium, gold and silver for their different parts.



Mixture vs. Compound - Iron and Sulfur

We compare behaviour of iron and sulfur when simply mixed and when they react together on heating. This classic classroom experiment highlights differences between mixtures and compounds.



Before heating - Sample A (mixture)

Mix iron filings and sulphur powder to make **Sample A**. This is a mixture because the two substances are combined without chemical change.



Appearance: You can see black iron filings and yellow sulphur powder separately-so it is non-uniform.

Magnet test: A magnet attracts only the iron filings, leaving the sulphur behind; each component retains its property.

Acid test: Add dilute hydrochloric acid to Sample A. The iron reacts to produce hydrogen gas (which makes a sharp "pop" sound when ignited); sulphur does not react and remains as yellow solid.

After heating - Sample B (compound)

Gently heating Sample A while stirring causes a chemical reaction that produces a new black solid called **iron sulphide** (Sample B). This is a compound.



Sample B

Appearance: The product is a uniform black mass; separate iron and sulphur are no longer visible.

Magnet test: A magnet does not attract Sample B, because iron is chemically combined and its magnetic behaviour changes.

Acid test: Adding dilute hydrochloric acid to Sample B releases hydrogen sulphide gas (with a rotten-egg smell), which is chemically different from the hydrogen gas evolved from the iron in the mixture.

Thus, compounds have fixed composition and properties different from their components; they cannot be separated into those components by simple physical methods.

Property	Mixture	Compound
Constituents	Two or more substances	Two or more elements
Separation	By physical methods (e.g., magnet, filtering)	By chemical methods only
Properties	Each component keeps its own	New properties, different from constituents
Ratio	Variable (can change proportions)	Fixed for each compound
Appearance	May be non-uniform or uniform	Always uniform (homogeneous)

How Do We Use Elements, Compounds and Mixtures?

Elements are the basic building blocks, compounds create new useful substances, and mixtures are tailored for practical uses. This understanding is central to innovations in health, food, construction and environmental science.

Everyday presence

Elements, compounds and mixtures surround us: Air (mixture), water (compound) and metals used in construction (elements or alloys) are examples.

Metals for construction: Elements such as iron and aluminium are used for bridges, buildings and vehicles because of strength and durability.

Role in science, technology and industry

New discoveries: Knowing how elements combine allows chemists and engineers to design new materials and medicines.

Medicines and vaccines: Chemists use compounds and reactions to develop life-saving drugs.

Fertilisers: Knowledge of compounds like nitrates and phosphates increases crop yield.

Engineering materials: Alloys and composite mixtures give improved strength, corrosion resistance and desired mechanical properties.

Building materials as mixtures

Many construction materials are mixtures: **wood, steel** and **concrete** are used for their combined properties.

Metals are extracted from minerals (which are naturally occurring mixtures or compounds) and processed to obtain elemental metals or alloys.



Graphene Aerogel

Special example: Graphene aerogel

What is it? A very lightweight material made from carbon (an element) and known as one of the lightest materials on Earth.

Properties: Highly porous with excellent absorbent capacity.

Uses: Environmental clean-up (absorbing oil spills), energy-saving devices, and special coatings for buildings.

What Are Minerals?

Minerals are naturally occurring substances found in rocks; most rocks are mixtures of different minerals.



Viewing minerals: Minerals can be seen with the naked eye, a magnifying glass or a microscope.

Building blocks: Minerals are either pure elements or compounds made of two or more elements.

Relation to matter: Elements and compounds in minerals are the basic building blocks of matter (anything with mass and volume).

Types of minerals

1. Native minerals: These occur in elemental form in nature. Examples include gold, silver, copper, sulphur and carbon (in forms like graphite or diamond).

2. Compound minerals: Most minerals are compounds made of two or more elements combined chemically. Examples: quartz, calcite, mica, olivine, talc.

Everyday uses of minerals

Minerals (or elements extracted from them) are used in many daily items due to their properties.

Examples:

Cement: Made from minerals like calcite, quartz, alumina, and iron oxide (or obtained from minerals); used in construction.

Talcum Powder: Made from the mineral talc; used for personal care.

Minerals provide raw materials for buildings, tools, and products, showing how elements and compounds from nature support human needs.

What Is Matter? (Concluding ideas)

Matter is anything that has mass and occupies space; it is built from elements and compounds and forms all material objects such as rocks, water and air.

Non-matter: Some things do not have mass or occupy volume in the same way; examples are light, heat, electricity, thoughts and emotions-important phenomena but not classified as matter.

Why it matters: Understanding matter and its forms helps explain everyday observations and enables technological and scientific progress.

You can practice questions from this chapter here: **Short and Long Answer Questions:**

Nature of Matter: Elements, Compounds, and Mixtures

(https://edurev.in/studytube/Short-and-Long-Answer-Questions-Nature-of-Matter-Elements--Compounds--and-Mixtures/8ee93a33-5fb4-4151-bf69-1a0a9aee8e51_t)